Research Plan

Quantum-Native Optimization for Risk-Sensitive URLLC in UAV-RIS Networks

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May 13, 2026

1. Project Title

Quantum-Native Joint Optimization of Codebook RIS, Multi-Stream NOMA Scheduling, and Mobility-Aware UAV Control for Risk-Sensitive URLLC: A Doubly-CVaR QAOA Framework

Domain: Quantum Optimization, Wireless Communication.

2. Project Overview

This project formulates and solves an optimization problem that maximizes the Energy Efficiency (EE) in next-generation wireless communication. Our work is to find a solution based on hybrid quantum-classical optimization approach. We consider an event-triggered downlink network in which unmanned aerial vehicle (UAV) base stations serve ground users through a segmented reconfigurable intelligent surface (RIS) under ultra-reliable low-latency communication (URLLC) constraints. The decision variables in the system are inherently discrete and jointly determined within a single optimization step.

Mobility enters the optimization in two ways. First, the proposed system formulation determines the predicted channel statistics over the upcoming event horizon via an Bernstein–Dinkelbach's process; these statistics flow into the QUBO coefficients. Second, the formulated binaries $z_{u,\ell}$ are themselves quantum decision variables, allowing the QAOA to trade exploration of users against channel coherence under different motion profiles.

2.1 Qubit allocation

A full waypoint-grid trajectory representation would require $O(UWH)$ qubits, where U, W, H denotes number of UAV, codeword information, height respectively. This exceeds NISQ budgets. The proposed template selection adds only $UL \sim 16$ qubits and keeps the airspace exploration combinatorial rather than fully continuous. Continuous velocity and acceleration within the chosen template are then refined classically.

We prove that the joint problem admits an exact Quadratic Unconstrained Binary Optimization (QUBO) formulation and design a hybrid quantum-classical algorithm in which QAOA [1,2] solves the discrete subproblem while a closed-form classical step allocates power and refines per-template velocity. The discreteness in our model reflects hardware reality [3,4], the combinatorial nature of scheduling and SIC ordering, and the strategic nature of mobility intent under URLLC constraints.

The reliability target is a Conditional Value-at-Risk (CVaR) constraint on Age of Information (AoI), capturing tail behavior of staleness rather than average performance. This is the *outer* use of CVaR: a system-level problem objective. We additionally adopt CVaR as the *inner* variational aggregator over QAOA measurement shots, following [5] for faster and more reliable convergence on NISQ hardware. The two CVaR uses operate on distinct distributions (Monte Carlo channel realizations for the outer; QAOA shot energies for the inner) and are mathematically compatible. Hence, we refer to the proposed framework as *doubly-CVaR optimization*.

The project delivers:

- A formal system model and problem formulation with four theoretical guarantees: exact QUBO equivalence, CVaR shot-complexity for QAOA, doubly-CVaR convergence equivalence, and hybrid Karush–Kuhn–Tucker (KKT) convergence under bounded template approximation.
- A modular Qiskit / PennyLane implementation of the doubly-CVaR hybrid solver, including the QUBO matrix builder, mobility template library, QAOA ansatz with CVaR-aggregated loss, and optimal post-processing algorithm.
- Benchmarking classical sub-optimal heuristic (like tabu search, matching, MPC trajectory + greedy scheduling) baselines against quantum NISQ simulation environment.
- An open-source repository documenting the QUBO encoding pipeline, intended to be reusable for other discrete wireless optimization problems (spectrum allocation, beam selection, slice scheduling, fleet routing).

The work positions quantum optimization as a alternate tool for solving NP Hard problems that are intrinsically combinatorial. It builds on the CVaR-augmented variational optimization line of [3] (which uses CVaR as a sample aggregator) and the recursive QAOA wireless work of [5] (which addresses interference-aware channel assignment), unifying and extending both into a URLLC-grade framework with mobility, reliability, and solvability guarantees.

3. Expected Pre-requisites and Preparation

- A solid grounding in linear algebra, probability, and basic quantum computing (qubits, gates, measurement, Bloch sphere).
- Familiarity with variational quantum algorithms at the level of QAOA and VQE; prior exposure to Ising / QUBO formulations is helpful.
- Working knowledge of Python and at least one quantum framework (Qiskit, PennyLane, or Qrisp). Experience with classical optimization (CVXPY, scipy.optimize) is a plus.
- A working understanding of wireless communication basics is helpful but not required; the relevant signal models will be covered in the first week.

4. Six-Week Execution Plan

Week	Milestones
Week 1	Understand the system model: codebook RIS, NOMA-SIC [6], the AoI metric [7], CVaR risk measures [8], and the Bernstein–Dinkelbach (B-D) mobility-coupled channel. Set up the Qiskit / PennyLane environment and reproduce a small QAOA MaxCut example with both expected-energy and CVaR shot aggregation [5].
Week 2	Derive the QUBO encoding for the joint RIS-codeword and user-scheduling subproblem. Implement the QUBO matrix builder that takes channel realizations and returns coefficients $\mathbf{Q}$, $\mathbf{q}$, q_0. Validate on a small instance ($K = 2$, $U = 1$, $N = 8$).
Week 3	Add SIC-order encoding via permutation binaries. Add mobility template selection binaries $z_{u,\ell}$ with a library of $L = 4$–8 pre-designed UAV templates (approach-cluster, hover, orbit-RIS, return-to-base). Implement constraint penalties. Verify QUBO equivalence on a medium instance via brute-force comparison.
Week 4	Integrate CVaR-AoI as the outer optimization objective. Implement Monte Carlo AoI sampling driven by the B-D channel model conditioned on each template's velocity profile. Aggregate steps from Week 1 to form the full doubly-CVaR loss surface.
Week 5	Complete the hybrid loop; classical inner smoother refines velocity within each selected template. Benchmark against classical MPC-trajectory + matching-scheduling, and standard expected-energy QAOA. Collect QAOA depth-vs-quality and inner-CVaR confidence-vs-quality curves.
Week 6	Optional: hardware run on IBM Eagle / Quantinuum H2. Scale to $\sim$160-qubit instances on Qiskit Aer / PennyLane simulators. Run robustness analysis under channel estimation error and template mismatch. Document, package, and write up results.

5. Deliverables

- A well-documented Jupyter notebook implementing the full hybrid quantum-classical pipeline: QUBO builder (codebook + scheduling + SIC + mobility templates), QAOA solver with CVaR shot aggregation, classical power allocation, classical velocity refinement, and outer CVaR-AoI evaluation. Submitted as an open-source contribution.
- A short technical report summarizing the system model, the QUBO derivation, the four theoretical guarantees (QUBO equivalence, shot-complexity, doubly-CVaR equivalence, hybrid KKT convergence), and benchmark results.
- (Optional) A full arXiv manuscript targeting IEEE JSAC or IEEE Transactions on Quantum Engineering, co-authored with interested members, presenting the framework as a standalone quantum-native contribution to wireless URLLC optimization.

6. Guidance and Extension

I will be actively involved in both the theoretical and implementation aspects of the project, providing regular support to address queries on the system model, the QUBO encoding, and the QAOA implementation. A short introductory session will be arranged in Week 1 to cover the wireless background and the quantum-native formulation. Weekly progress meetings will track milestones, with additional meetings on request.

As for extending the work beyond the project duration, we can collaboratively pursue: (i) hardware validation on near-term quantum processors, (ii) replacement of QAOA by alternative ansatz families (QAOA+ [9], adaptive mixers) for tighter constraint preservation, (iii) comparison against quantum annealing on D-Wave hardware.

References

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